Taxation of Couples: Payroll Tax and Transferable Benefits

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Abstract

This paper studies the optimal design of payroll taxes when couples jointly choose between formal and informal work and social insurance benefits are transferable: a spouse can retain coverage (e.g., health insurance) through their partner's formal job. Transferability gives households an incentive to specialize: one spouse provides coverage formally while the other works informally. I develop a model in which the government sets both a payroll tax rate and a transferability parameter, and derive optimal policy formulas.

Higher transferability has two opposing effects on the elasticity of the formal/informal choice with respect to the tax rate. In one-formal households, the sole provider becomes less elastic, since more coverage is at stake from exiting formality (the provider-role channel). In both-formal households, the second spouse becomes more elastic, since less is at stake (the second-spouse channel). Which channel dominates depends on labor-market structure: the provider-role channel dominates under substantial informality, while the second-spouse channel dominates in fully-formal economies, raising the efficiency cost and lowering the optimal tax. I test the mechanism using Chile's Bono al Trabajo de la Mujer, a payroll tax subsidy for low-income women with full spousal transferability. Eligible women increased formal participation — especially those married to informal men — while their spouses shifted toward informal work, consistent with strategic substitution within households. This places Chile in the provider-role regime, where greater transferability raises the optimal payroll tax.

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1. Introduction

Social insurance and payroll tax systems have historically been designed around single-earner households. This means that workers are subject to payroll in exchange for benefits such as pensions or health insurance for themselves. However, in modern dual-earner households, benefits are often transferable to dependents or spouses. This shift may encourage new behavioral responses in the labor market: a worker can participate in the labor market informally (outside the taxed formal sector), avoid payroll taxes, yet still enjoy coverage through their partner's formal employment. As a result, the optimal design of payroll taxes and social insurance may need rethinking to account for these intra-household spillovers. In particular, policymakers face a tension between providing insurance to workers' families and preventing strategic informality, where one spouse remains informal to avoid taxes while retaining coverage through the other's formal employment.

These issues are especially acute in middle-income and developing countries, where informality is very high. Despite numerous formalization efforts, a large share of the workforce in developing economies continues to operate informally. For example, in Latin America, between 30 and 70% of workers work without being enrolled in the payroll system nor subject to any taxation. These high informality rates have proven persistent over time, reflecting deep-rooted social norms and institutional settings.

Workers weigh the costs of payroll taxes against the value of benefits; when benefits can be obtained without formal participation (for instance, via a spouse), the benefit to formal employment decreases. Informality can thus be viewed as an outcome of optimal behavior by workers given the institutional settings. This perspective aligns with evidence that household context plays a crucial role in labor supply decisions. If one spouse's formal job secures the household's social insurance, the other spouse may find it optimal to remain untaxed and informal. Such intra-household coordination of formal and informal work presents a novel challenge for tax design: there is a disconnection between who pays for social insurance and who ultimately benefits.

Motivated by this context, I develop a partial-equilibrium model in which couples choose formality jointly, and the government selects both a payroll tax rate and a benefit transferability parameter to maximize welfare. Each household consists of two spouses who jointly decide whether to work in the formal sector (paying payroll taxes and receiving benefits for themselves and the spouse) or in the informal sector (avoiding taxes, foregoing benefits, but potentially receiving them from the spouse). A key feature of the model is a transferability parameter ($0 \leq \gamma \leq 1$) that represents what fraction of a formal worker's benefits can be transferred to their spouse. At one extreme, $\gamma = 0$ represents a system with no spousal benefit sharing; at the other extreme, $\gamma = 1$ represents fully transferable benefits (a one-earner couple can cover both members). The government chooses two policy instruments: (i) a payroll tax rate (τ) on formal employment income, and (ii) the degree of benefit transferability (γ).

To build intuition, consider first the case of no transferability ($\gamma = 0$). Decisions are individual: each worker's formality decision depends only on their own (payroll) tax cost and own benefit. Decisions are separable across spouses, there is no cross-spousal interaction, and the optimal tax reduces to the standard condition, depending only on own elasticities of informality with respect to the net-of-tax rate.

Now fix transferability at a positive level and ask how the optimal tax changes. The optimal tax 'formula' shows that the sufficient statistic for the optimal payroll tax will be the elasticity of informality

with respect to the net-of-tax rate, $(1-\tau)$, which is negative (higher net-of-tax rate, lower informality). When transferability is positive, the couple has incentives to specialize: one spouse serves as the coverage provider for the household.

This specialization generates two opposing effects on household elasticities. The provider-role channel: the sole provider in a one-formal household becomes *less* responsive, since exiting would cost the household $(1+\gamma)b$ in coverage, the full benefit. The second-spouse channel: the second spouse in a both-formal household becomes *more* responsive, since exiting formality costs the household only $(1-\gamma)b$ in coverage, the partner still provides γb. The model shows that the net effect on the optimal payroll tax depends on which margin carries more mass in the population: the both-formal (FF) margin or the one-formal (FI/IF) margin. In economies where one-formal households are common (e.g., middle-income economies with substantial informality), the provider-role channel dominates, the elasticity-weighted efficiency cost shrinks with transferability, and the optimal payroll tax is *higher* than in the no-transferability benchmark. In fully-formal economies where most couples are fully formal, the prediction reverses: the second-spouse channel dominates, transferability amplifies the elasticity, and the optimal tax is *lower*. The model's prediction therefore depends on the labor-market structure, not on the level of transferability alone.

The analysis of the tax rate assumes a fixed degree of transferability. However, transferability is not a fixed parameter, it is itself a policy instrument. The 'formula' shows that the sufficient statistic for the optimal transferability is the elasticity of informality with respect to the degree of transferability (γ), whose sign is ambiguous. And the sign of the elasticity determines the optimal policy. Higher transferability strengthens the provider role: the coverage the provider brings becomes more valuable and the second spouse's contribution becomes even less necessary. This pushes toward one-formal arrangements from both extremes (both-formal and both-informal), so in one case it increases informality and in the other it decreases it.[1] If the both-formal to one-formal margin dominates (more informality), the optimal 'formula' resembles the tax one: there is a trade-off between the efficiency cost of increasing transferability (more informality) and the gain from extending coverage to partners. The sufficient statistic is the elasticity, and the optimal transferability depends on its magnitude. If the both-informal to one-formal margin dominates (less informality), the increase in transferability generates a net decrease in informality, there is no trade-off and full transferability is optimal.

I test the mechanism using Chile's Bono al Trabajo de la Mujer (BTM), a payroll tax subsidy for low-income women. Chile is a natural setting: roughly 30% of the working-age population is informal, one-formal households are a substantial share of the marginal mass, and benefits are fully transferable across spouses. Eligible women significantly increased formal participation, especially those married to informal men, while their spouses became more likely to work informally. These patterns are consistent with strategic substitution: as the wife formalizes, the husband's coverage is secured, reducing his incentive to remain formal. These estimates speak to the sufficient statistics for the optimal tax. Because transferability does not vary in Chile, they do not identify the elasticities that govern the optimal transferability.

[1] In both-formal households, the second spouse exits because the provider already secures enough coverage. In both-informal households, one spouse enters formality because the provider role is now valuable enough to justify the tax cost. One-formal households are relatively unaffected at the margin (they won't change their behavior).

This study contributes to the literature on the taxation of couples and multi-person households (e.g., Saez 2002; Kleven, Kreiner, and Saez 2009; Golosov and Krasikov 2025). Previous models of couple taxation have typically assumed a fully joint tax schedule or solved a multi-dimensional screening problem to design optimal taxes based on both spouses' incomes. For example, Kleven et al. (2009) model couples as a unitary household with a primary and a secondary earner, and show that the optimal tax system features complex jointness—the tax rate on one spouse may depend on the earnings of the other. Similarly, recent mechanism-design approaches derive optimal nonlinear taxes for couples, allowing for different marginal rates across spouses and accounting for assortative matching (Golosov and Krasikov 2025).

While powerful, existing models are highly complex and often difficult to implement in practice. In contrast, this paper focuses on individual taxation with spillovers. It does not require the government to set fully separate tax schedules for couples or to observe the joint distribution of earnings. Instead, household interactions are captured through the single transferability parameter (γ) in the benefit formula. This gives a tractable framework that still reflects key policy questions surrounding couples' labor supply decisions. By solving for the optimal payroll tax and transferability parameter, the study offers an alternative lens on couple taxation: rather than redesigning the entire tax schedule for two earners, the government can adjust the degree of benefit transferability to address inefficiencies arising from intra-household dynamics. This simpler policy lever could be more feasible in practice (for instance, adjusting whether non-contributing spouses are eligible for full, partial, or no benefits).

This study also relates to the literature on informality in developing economies (e.g., Maloney 2004; Loayza 2018; Ulyssea 2020). A rich body of research examines why informal employment remains high and how it responds to taxes, regulations, and enforcement. However, much of this work analyzes individuals or firms in isolation. Here, the household is centered as the decision-making unit, highlighting a new channel for informality.

This paper shows that inter-spousal incentives can distort labor supply: a worker's decision to enter or stay in the formal sector depends not only on their own costs and benefits, but also on whether their partner is already formal and can share benefits. This mechanism helps explain observed patterns such as couples rarely both working informally. It also suggests that conventional policies to reduce informality (such as tax incentives) might have unintended consequences unless they account for family ties. This paper highlights one specific policy margin—benefit transferability—that can either mitigate or exacerbate informality.

Finally, this paper relates to the literature on how tax systems shape household structure itself (Gayle and Shephard, 2019; Chade and Ventura, 2002; Persson, 2020). In that literature, the tax schedule affects who marries whom and how couples allocate time, so the marriage market is endogenous to tax policy. Here I take household composition as given and derive optimal policy conditional on the labour-market structure. Endogenizing household formation—allowing the tax and transferability instruments to reshape the distribution of couple types—is a natural extension of the framework.

Overall, this work bridges the gap between traditional individual-based tax theory and the reality of households in middle-income economies, providing guidance for how to tailor payroll tax policies when one worker's benefits effectively extend to two people. The remainder of the paper formalizes the model, characterizes the optimal payroll tax and transferability parameter, and discusses the results and their

implications for policy and future research.

2. Model

This is a partial-equilibrium model that shows how transferable payroll benefits and a payroll tax can generate spillovers in spouses' formality decisions, with implications for optimal policy design.

2.1. Set-Up

Consider an economy with N married men (m) and women (f), normalized such that $N = 1$. No one remains single, and marriages are exogenously formed through random matching. Each individual $i \in m, f$ differs along one dimension: a stigma cost of informality, c_i. This cost reflects legal or social repercussions associated with informal employment. For men and women, these costs are drawn from gender-specific distributions $q_m(c_i^m)$ and $q_f(c_i^f)$, with corresponding cumulative distribution functions F_m and F_f. These costs are drawn independently across individuals, with no within-couple correlation between c_m and c_f[2].

There is a linear payroll tax (τ), levied only on formal employment. Paying payroll taxes gives individuals access to payroll benefits (b), which enter their utility. The economy has no income tax. Payroll benefits are partially transferable between spouses: if one spouse has a formal job (and thus pays payroll taxes and receives benefits), the other receives a share of those benefits, γb. Both τ and γ are policy instruments chosen by the planner. Finally, there is a subsidy s for women's formal employment that reduces their effective payroll tax rate. I assume this subsidy is exogenously given. It allows for differentiated effective payroll tax rates between men and women.

The main assumptions are as follows: (i) wages (z_m, z_f) are exogenous; (ii) although informality costs are independently distributed across spouses, employment decisions are made jointly to maximize household utility; and (iii) the subsidy s is exogenous.

2.2. Workers

Household Utility Individuals are fully employed, and their only labor-supply margin is whether to work formally or informally. For spouse i, pre-tax income is z_i, where z_m denotes male wages and z_f denotes female wages. Wages are assumed to be constant across formal and informal jobs and differ only by gender.

Formal workers are subject to a linear payroll tax τ (reduced by a subsidy s for women) and gain access to payroll benefits b. Their spouse also receives a share γb of those benefits. Thus, the utility of a formal male worker is:

$$u_m^F = (1 - \tau)z_m + b \tag{1}$$

For a formal female worker:

$$u_f^F = [1 - (1 - s)\tau]z_f + b \tag{2}$$

[2]Assuming independent stigma costs abstracts from sorting in the marriage market but simplifies the analysis of joint household decisions. Introducing assortative matching or correlated costs could yield richer predictions about how couples sort into formality based on shared attitudes toward informality.

Informal workers avoid payroll taxes but incur a stigma cost c_i. If their spouse is formally employed, they still receive a share γb of the benefits. Hence, the utility of an informal worker i with spouse j is:

$$u_i^I = z_i - c_i + \Big(1 - \mathbb{1}\{j \text{ is informal}\}\Big)\gamma b \tag{3}$$

where the indicator $\mathbb{1}\{j \text{ is informal}\} = 1$ if spouse j is informal and 0 otherwise.

The total household utility is defined as the sum of the male and female individual utilities:

$$U^{\text{HH}} = u_m + u_f, \tag{4}$$

Each household (couple) jointly chooses among four possible outcomes:

1. **FF:** Both spouses are formal.
2. **FI:** Husband is formal and wife is informal.
3. **IF:** Husband is informal and wife is formal.
4. **II:** Both spouses are informal.

The household utility under each regime is given by:

$$U^{\text{FF}} = (1-\tau)z_m + (1-(1-s)\tau)z_f + 2b, \tag{5}$$
$$U^{\text{FI}} = z_m(1-\tau) + b + z_f + \gamma b - c_f, \tag{6}$$
$$U^{\text{IF}} = z_m + \gamma b - c_m + (1-(1-s)\tau)z_f + b, \tag{7}$$
$$U^{\text{II}} = z_m + z_f - (c_m + c_f). \tag{8}$$

Households choose the regime that yields the highest U^{HH} among the four options. The indifference conditions between regimes are derived by equating the corresponding utility expressions[3]:

$$\begin{aligned}
\text{(FF vs. FI):} \quad & c_f = (1-s)\tau z_f - (1-\gamma)b, \\
\text{(FF vs. IF):} \quad & c_m = \tau z_m - (1-\gamma)b, \\
\text{(FI vs. II):} \quad & c_m = \tau z_m - (1+\gamma)b, \\
\text{(IF vs. II):} \quad & c_f = (1-s)\tau z_f - (1+\gamma)b, \\
\text{(FI vs. IF):} \quad & c_m - c_f = \tau[z_m - (1-s)z_f]
\end{aligned}$$

These indifference conditions characterize the household's regime choice in the (c_m, c_f) space. After comparing all boundaries, one can identify that each spouse faces two distinct thresholds, depending on whether the partner is formal or informal. To simplify notation, define the thresholds implied by these

[3]The FF vs. II comparison is omitted because it is implied by the other conditions. Setting $U^{\text{FF}} \geq U^{\text{II}}$ yields $c_m + c_f \geq \tau z_m + (1-s)\tau z_f - 2b$. When the FF vs. FI and FF vs. IF conditions both hold ($c_m \geq T_m$ and $c_f \geq T_f$), we have $c_m + c_f \geq T_m + T_f = \tau z_m + (1-s)\tau z_f - 2(1-\gamma)b \geq \tau z_m + (1-s)\tau z_f - 2b$, where the last inequality follows from $\gamma \geq 0$.

indifference conditions as

$$T_m = \tau z_m - (1-\gamma)b, \qquad T_m^{\text{low}} = \tau z_m - (1+\gamma)b, \tag{5}$$

$$T_f = (1-s)\tau z_f - (1-\gamma)b, \qquad T_f^{\text{low}} = (1-s)\tau z_f - (1+\gamma)b. \tag{6}$$

For each spouse $i \in \{m, f\}$, these thresholds satisfy $T_i - T_i^{\text{low}} = 2\gamma b$, because the gain from formalizing is larger when the spouse is informal than when the spouse is already formal. That is, the cost cutoff for formality is higher when the spouse is already formal (e.g., T_m, the threshold for male formality when the wife is already formal, is higher than T_m^{low}, the threshold when she is informal).

The indifference conditions imply that the decision regions in the (c_m, c_f) space are not fully separable. While the fully formal and fully informal regions are rectangular, the one-formal regions depend jointly on both spouses' costs and are separated by the diagonal condition above in Eq. (9).

Specifically, a household chooses **FF** if

$$c_m \geq T_m \quad \text{and} \quad c_f \geq T_f.$$

A household chooses **II** if

$$c_m < T_m^{\text{low}} \quad \text{and} \quad c_f < T_f^{\text{low}}.$$

All remaining combinations of (c_m, c_f) correspond to households with exactly one formal spouse. Within this set, the household prefers having one formal worker, but which spouse is formal depends jointly on both spouses' costs. The boundary between **FI** and **IF** is given by

$$c_m - c_f \;=\; \tau\big[z_m - (1-s)z_f\big] \;\equiv\; D. \tag{7}$$

I denote this diagonal by D. Under the assumption $z_m \geq (1-s)z_f$ (husband's gross wage at least as large as wife's after-subsidy wage), $\partial D/\partial \tau \geq 0$ and $D \geq 0$. The one-formal regions are separated by this diagonal and cannot be represented as rectangular sets.

Figure 1 illustrates the geometry. The four thresholds $T_m^{\text{low}}, T_m, T_f^{\text{low}}, T_f$ partition (c_m, c_f) space into a 3×3 grid; FF (top-right) and II (bottom-left) are rectangular and factorise, while FI and IF each consist of two rectangular strips plus a triangular piece in the central *intermediate rectangle* $[T_m^{\text{low}}, T_m] \times [T_f^{\text{low}}, T_f]$.

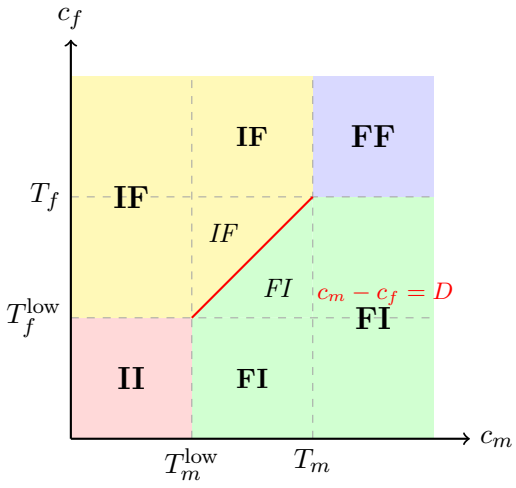
Because c_m and c_f are independently distributed, the probabilities of the fully formal and fully informal regimes factorize:

$$P_{\text{FF}} = \left[1 - Q_m(T_m)\right]\left[1 - Q_f(T_f)\right], \tag{8}$$

$$P_{\text{II}} = F_m\big(T_m^{\text{low}}\big) F_f\big(T_f^{\text{low}}\big). \tag{9}$$

In contrast, the probabilities of the one-formal regimes do not factorize, because in the central rectangle the allocation of formality depends jointly on both spouses' costs through the diagonal condition.

Figure 1: Household decision regions in (c_m, c_f) stigma-cost space.



For example, the probability of the **FI** outcome can be written as

$$P_{\text{FI}} = [1 - Q_m(T_m)]\, Q_f(T_f) \tag{10}$$

$$+ \left[Q_m(T_m) - F_m\big(T_m^{\text{low}}\big)\right] F_f\big(T_f^{\text{low}}\big) \tag{11}$$

$$+ \int_{\mathcal{D}_{FI}} q_m(c_m) q_f(c_f)\, dc_m\, dc_f, \tag{12}$$

where $\mathcal{D}_{FI}$ denotes the region where both costs lie in the intermediate range $T_m^{\text{low}} \leq c_m < T_m$, and $T_f^{\text{low}} \leq c_f < T_f$, and where the household prefers the husband to be the formal worker, that is $c_m - c_f \geq \tau\big[z_m - (1-s)z_f\big]$. The probability P_{IF} is defined symmetrically.

Probability that spouse i is informal. The object that enters the planner's elasticities is the probability that spouse i is informal. This is the sum of the two regimes in which i is informal:

$$F_m \;\equiv\; P_{\text{II}} + P_{\text{IF}}, \qquad F_f \;\equiv\; P_{\text{II}} + P_{\text{FI}} \tag{13}$$

At $\gamma = 0$ (the no transferability case) the central rectangle in Figure 1 collapses and F_i coincides with $Q_i(T_i)$ (the CDF of stigma costs evaluated at i's threshold). For any positive level of transferability, $\gamma > 0$, the two need not be the same and $F_m F_f \neq P_{\text{II}}$ because the marginal informality probabilities are not independent.

2.3. Planner

I assume the planner maximizes a generalized social welfare function. The planner chooses the linear payroll tax τ (and associated benefit b) and the level of transferability of payroll benefits γ to maximize welfare. Thus, the social welfare function is defined as:

$$\begin{aligned}\mathrm{SWF}(\tau,\gamma) &= P_{\mathrm{FF}}\; G(U^{\mathrm{FF}}) + P_{\mathrm{FI}}\; G(U^{\mathrm{FI}}) + P_{\mathrm{IF}}\; G(U^{\mathrm{IF}}) + P_{\mathrm{II}}\; G(U^{\mathrm{II}}) \\ &= \;\; P_{\mathrm{FF}}\; G(u_m^F + u_f^F) + P_{\mathrm{FI}}\; G(u_m^F + u_f^I) \\ &\quad + P_{\mathrm{IF}}\; G(u_m^I + u_f^F) + P_{\mathrm{II}}\; G(u_m^I + u_f^I)\end{aligned} \tag{14}$$

where $G(\cdot)$ is an increasing, concave function (e.g., log or CRRA) representing the government's redistributive preferences. The planner evaluates welfare at the household level, considering total household utility rather than the utilities of individual spouses.

The government collects revenue through a payroll tax: men pay a rate τ, while women pay a reduced rate $(1-s)\tau$ due to the subsidy. In a couple with outcome (FF), both spouses contribute: the man pays τz_m and the woman pays $(1-s)\tau z_f$. In a (FI) household, only the man pays τz_m, and in a (IF) household, only the woman pays $(1-s)\tau z_f$. Formally, I define total revenue (R) as the total income subject to tax (Z) times the tax rate τ:

$$R = \tau\Big[\underbrace{(z_m + (1-s)z_f), P_{\mathrm{FF}}}_{\text{both pay}} + \underbrace{z_m\; P_{\mathrm{FI}}}_{\text{only } m \text{ pays}} + \underbrace{(1-s)z_f\; P_{\mathrm{IF}}}_{\text{only } f \text{ pays}}\Big] = \tau Z \tag{15}$$

In this economy, payroll benefits are partially transferable between spouses. Any household with at least one formal worker receives full coverage for one spouse and partial coverage (a share γ) for the other. Thus, households in (FF) receive $2b$, while those in (FI) and (IF) each receive $(1+\gamma)b$. Total government expenditure on benefits is:

$$\text{Expenditure}(\tau) = b\big[2P_{\mathrm{FF}} + (1+\gamma)(P_{\mathrm{FI}} + P_{\mathrm{IF}})\big] = bN \tag{16}$$

where N is the total benefit obligation (weighted by how much benefit each recipient gets).

Imposing a balanced-budget condition, one obtains the following government budget constraint:

$$R(\tau,\gamma) = \text{Expenditure}(\tau,\gamma) \tag{17}$$

The planner chooses τ and γ to maximize:

$$\mathcal{L}(\tau,\gamma,\lambda) \;=\; \mathrm{SWF}(\tau,\gamma) + \lambda\big[R(\tau,\gamma) - \text{Expenditure}(\tau,\gamma)\big] \tag{18}$$

Differentiating with respect to τ and setting $\mathrm{d}\mathcal{L}/\mathrm{d}\tau = 0$ yields the optimal tax condition:

$$\frac{\mathrm{d}\,\mathrm{SWF}(\tau,\gamma)}{\mathrm{d}\tau} \;+\; \lambda\frac{\mathrm{d}}{\mathrm{d}\tau}\Big[R(\tau,\gamma) - \text{Expenditure}(\tau,\gamma)\Big] = 0 \tag{19}$$

Similarly, differentiating with respect to γ and setting $\mathrm{d}\mathcal{L}/\mathrm{d}\gamma = 0$ yields the optimal transferability condition:

$$\frac{\mathrm{d}\,\mathrm{SWF}(\tau,\gamma)}{\mathrm{d}\gamma} \;+\; \lambda\frac{\mathrm{d}}{\mathrm{d}\gamma}\Big[R(\tau,\gamma) - \text{Expenditure}(\tau,\gamma)\Big] = 0 \tag{20}$$

3. Optimal Payroll Tax and Transferability

3.1. Key terms

Before getting into the optimal formulas, here I define the following elasticities of marginal informality F_i, introduced in equation (13), for $i \in \{m, f\}$:

$$\epsilon_i = \frac{\partial F_i}{\partial(1-\tau)} \cdot \frac{1-\tau}{F_i}, \qquad \eta_i = \frac{\partial F_i}{\partial \gamma} \cdot \frac{\gamma}{F_i}. \tag{21}$$

The terms ϵ_m and ϵ_f represent the extensive-margin elasticities of informality with respect to the net-of-tax rate, while η_m and η_f capture the elasticities with respect to the transferability parameter γ.

ϵ_i captures the change in the probability of informality of spouse i when there is a change in the net-of-tax rate. Take the husband as example. There are three potential flows that could happen that increase his probability of being an informal worker: the switches from FF to IF (his own threshold T_m shifts as τ rises), from FI to II (his own threshold T_m^{low} shifts), and from FI to IF (the diagonal D shifts, and the household switches which spouse is the formal earner).[4]

To separate the channels—those related to their own thresholds and the one related to the diagonal—I decompose the total elasticity ϵ_i:

$$\epsilon_i \;=\; \epsilon_i^{\text{own}} + \epsilon_i^{\text{diag}}, \tag{22}$$

On one side, ϵ_i^{own} captures the first two switches (for the husband, for example, from FF to IF and from FI to II), those that happen across i's own thresholds T_i and T_i^{low}. These always increase i's informality, thus $\epsilon_i^{\text{own}} < 0$. On the other side, ϵ_i^{diag} captures the FI→IF reallocation across the diagonal D. By construction, $\epsilon_m^{\text{diag}} < 0$ (the diagonal pushes the husband further into informality) and $\epsilon_f^{\text{diag}} > 0$ (it pulls the wife out). I assume $|\epsilon_i^{\text{diag}}|$ is small relative to $|\epsilon_i^{\text{own}}|$. The diagonal channel involves only couples sitting close to the FI–IF boundary, whereas the own-threshold channel involves couples crossing any of the four threshold lines (typically a much larger population). For the rest of the paper I treat ϵ_i^{diag} as negligible and write $\epsilon_i = \epsilon_i^{\text{own}}$.

Turning to the transferability elasticity: the sign of η_i is ambiguous. Higher transferability operates on two margins. In a both-formal household, the second spouse's gain from being formal is only $(1-\gamma)b$; raising γ reduces this gain, increasing informality on the FF margin. In a both-informal household, the first spouse's gain from formalizing is $(1+\gamma)b - \tau z_i$; raising γ increases this gain, decreasing informality on the II margin. The net effect depends on which margin has more mass of households.

[4] For the wife, the picture is parallel but the diagonal piece has the opposite sign. Two of her own-threshold flows increase her informality: the switches from FF to FI and from IF to II. The third flow—from FI to IF—decreases her informality: she was informal in FI and becomes formal in IF. The same diagonal flow pushes the husband further into informality and pulls the wife out of it. A natural question is whether the reverse flow (IF to FI) can also occur, i.e., whether some couples could go from "wife is the formal earner" back to "husband is the formal earner" when τ rises. Under the natural sign convention $z_m \geq (1-s)z_f$, the diagonal $D = \tau(z_m - (1-s)z_f)$ increases with τ. The locus $c_m - c_f = D$ shifts in such a way that couples just on the FI side of the old diagonal (with $c_m - c_f$ just above the old D) cross over to the IF side (their $c_m - c_f$ is now just below the new, larger D). Couples on the IF side stay on the IF side. So raising τ produces FI-to-IF flips but never IF-to-FI flips. The reverse direction would require D to decrease (either τ falling, or a reversed effective-wage ranking $(1-s)z_f > z_m$).

Then, I define the ratios α_i and β_i as follows:

$$\alpha_m = \frac{z_m F_m}{Z} \qquad \alpha_f = \frac{(1-s) z_f F_f}{Z} \tag{23}$$

$$\beta_i \;=\; \frac{F_i(1-\gamma+2\gamma\,\mu_i)}{N} \tag{24}$$

α_i is the ratio of informal income earned by spouse i to the formal tax base Z. The numerator $z_i F_i$ measures the income spouse i earns while informal; the denominator Z is the current formal tax base. α_i captures the size of the informal sector for spouse i relative to the formal sector.

β_i is the ratio of expected coverage lost per unit of spouse-i informality, to total benefit obligations. The numerator $F_i(1-\gamma+2\gamma\mu_i)$ aggregates two types of coverage losses: when the exiting spouse was married to a formal partner, the household drops from full coverage $2b$ to partial coverage $(1+\gamma)b$ (loss $1-\gamma$); when married to an informal partner, the household drops from partial coverage $(1+\gamma)b$ to zero (loss $1+\gamma$). The share of spouse i's exiters whose partner is also informal is denoted μ_i; the complementary share $(1-\mu_i)$ are those married to a formal partner.[5]

Finally, let λ be the budget constraint multiplier. I define the following average social welfare weights at the household level:

$$g^{\mathrm{FF}} = \frac{G'(U^{\mathrm{FF}})}{\lambda}, \qquad g^{\mathrm{FI}} = \frac{\iint_{\mathcal{R}_{\mathrm{FI}}} G'(U^{\mathrm{FI}}) q_m(c_m) q_f(c_f)\, dc_m\, dc_f}{P_{\mathrm{FI}}\lambda} \tag{25}$$

$$g^{\mathrm{IF}} = \frac{\iint_{\mathcal{R}_{\mathrm{IF}}} G'(U^{\mathrm{IF}}) q_m(c_m) q_f(c_f)\, dc_m\, dc_f}{P_{\mathrm{IF}}\lambda}, \qquad g^{\mathrm{II}} = \frac{\int_0^{T_m^{\mathrm{low}}}\int_0^{T_f^{\mathrm{low}}} G'(U^{\mathrm{II}}) q_m(c_m) q_f(c_f)\, dc_f\, dc_m}{P_{\mathrm{II}}\lambda}$$

where $\mathcal{R}_{\mathrm{FI}}$ and $\mathcal{R}_{\mathrm{IF}}$ denote the (non-rectangular) decision regions for the one-formal regimes as characterized in Section 2. The II region is rectangular with bounds $c_m < T_m^{\mathrm{low}}$ and $c_f < T_f^{\mathrm{low}}$, while the FF region does not depend on (c_m, c_f).

3.2. Optimal τ

The first-order condition for τ equates the costs and benefits of increasing the tax rate:

$$\underbrace{-Z\bar{g}}_{\text{cost to taxpayers}} \;+\; \underbrace{\frac{db}{d\tau}\,\bar{N}}_{\text{benefit from increased } b} \;=\; 0, \tag{26}$$

$$\bar{N} \;\equiv\; 2P_{\mathrm{FF}}\, g^{\mathrm{FF}} + (1+\gamma)(P_{\mathrm{FI}}\, g^{\mathrm{FI}} + P_{\mathrm{IF}}\, g^{\mathrm{IF}}) \;=\; N + \Sigma,$$

$$\Sigma \;\equiv\; 2P_{\mathrm{FF}}(g^{\mathrm{FF}}-1) + (1+\gamma)\big(P_{\mathrm{FI}}(g^{\mathrm{FI}}-1) + P_{\mathrm{IF}}(g^{\mathrm{IF}}-1)\big),$$

The FOC shows that an increase in τ has two welfare effects, both weighted by the planner's social welfare weights. First, formal workers pay more in taxes: each additional unit of revenue $Z\,d\tau$ comes out of formal income, valued at the average welfare weight $\bar{g} \;=\; g^{\mathrm{FF}}\,\frac{P_{\mathrm{FF}}(z_m+(1-s)z_f)}{Z} \;+\; g^{\mathrm{FI}}\,\frac{P_{\mathrm{FI}}\,z_m}{Z} \;+$

[5] Formally, μ_i is the share of spouse i's own-threshold mass flow that occurs at the II margin . For the husband: $\mu_m = \frac{q_m(T_m^{\mathrm{low}})\,Q_f(T_f^{\mathrm{low}})}{q_m(T_m)\,[1-Q_f(T_f)] + q_m(T_m^{\mathrm{low}})\,Q_f(T_f^{\mathrm{low}})}$, with μ_f defined analogously. At $\gamma = 0$, $\mu_i = Q_j(T_j) = F_j$ (the partner's unconditional informality probability)

$g^{\text{IF}} \frac{P_{\text{IF}}(1-s)z_f}{Z}$ on formal workers. This is a cost to taxpayers, equivalent to $Z\,\bar{g}\,d\tau$. Second, the higher tax revenue is redistributed via a higher per-recipient benefit b: each recipient gains $(db/d\tau)\,d\tau$ in dollars. Each dollar of benefit is valued at the recipient's welfare weight, so the welfare gain is $(db/d\tau)\,\bar{N}\,d\tau$, where $\bar{N}$ is the welfare-weighted count of beneficiaries. The full derivation from the Lagrangian is in Appendix A.

Proposition 1 (Optimal payroll tax)**.** *Under the assumptions of Section 2, the optimal payroll tax rate that maximizes social welfare subject to the budget constraint $\tau Z = bN$ is*

$$\tau^* \;=\; \frac{1 + \frac{\Sigma}{N} - \bar{g}}{1 + \frac{\Sigma}{N} - \bar{g} \;-\; \left(1 + \frac{\Sigma}{N}\right)\left[\epsilon_m(\alpha_m - \beta_m) + \epsilon_f(\alpha_f - \beta_f)\right]} \tag{27}$$

where ϵ_i is the elasticity of spouse i's informality with respect to the net-of-tax rate, α_i is the informal income ratio, $\beta_i = F_i(1-\gamma+2\gamma\mu_i)/N$ is the coverage-loss ratio, $\bar{g}$ is the income-weighted average welfare weight of formal taxpayers, and Σ/N is the per-recipient net social value of benefits.

Equation (27) represents the standard equity-efficiency trade-off. The numerator captures the welfare gains of collecting one additional dollar of revenue and using it to fund more b. $1 + \frac{\Sigma}{N} = \frac{\bar{N}}{N}$ represents the average social value of of an additional dollar of benefit b. The '1' part is the mechanical part of \$1 and $\frac{\Sigma}{N}$ captures the marginal benefit (or cost) of receiving getting b instead of \$1.[6] When benefits are more valuable per recipient, it is optimal to have a higher tax rate to fund a more socially valuable redistribution.[7]

The denominator captures the efficiency cost of taxation. The term $\epsilon_i(\alpha_i - \beta_i)$ captures the effect for spouse i from exiting formality. $\epsilon_i < 0$ is the elasticity of spouse i's informality with respect to the net-of-tax rate. It is negative because higher net-of-tax rates decrease informality. Larger (in absolute value) elasticities imply greater efficiency losses, lowering the optimal tax rate. The ratio α_i measures informal income relative to the formal tax base (the revenue channel); a higher α_i raises the efficiency cost and reduces τ^*. The ratio β_i measures expected coverage lost per unit of informality relative to total benefit obligations (the coverage channel); a higher β_i means that a marginal switch to informality destroys proportionally more coverage and raises b for the remaining covered individuals, decreasing the social cost of distortions and thereby raising the optimal tax rate.

Comparative static of τ^* in γ. The formula gives τ^* for a fixed γ. Here I ask how τ^* changes when γ is varied, holding the primitives of the economy fixed.

Proposition 2 (Effect of transferability on τ^*)**.** *Let $\tau^*(\gamma)$ denote the solution to the τ-FOC (26) for a given γ, holding wages (z_m, z_f), stigma distributions (q_m, q_f), and the female subsidy s fixed. Differentiating (27) in γ:*

$$\frac{\partial \tau^*}{\partial \gamma} \;=\; \frac{1}{\text{den}^2}\Big[\,\text{num}\left(1 + \frac{\Sigma}{N}\right) E' \;-\; \bar{g}\,E\left(\frac{\Sigma}{N}\right)' \;+\; \left(1 + \frac{\Sigma}{N}\right) E\,\bar{g}'\,\Big], \tag{28}$$

[6] I use the $1 + \Sigma/N$ form throughout because Σ/N has the clean interpretation as the per-recipient net social value of an additional dollar of benefit.

[7] Differentiating (27) with respect to Σ/N: $\frac{d\tau^*}{d(\Sigma/N)} = \frac{\bar{g}\,|E|}{\text{den}^2} > 0$, where E and den are as defined in Proposition 2.

where $\text{num} = 1 - \bar{g} + \frac{\Sigma}{N} > 0$ *is the numerator of* τ^**,* $E \equiv \epsilon_m(\alpha_m - \beta_m) + \epsilon_f(\alpha_f - \beta_f) < 0$ *is the elasticity-weighted efficiency cost,* $\text{den} = \text{num} - (1 + \frac{\Sigma}{N})\, E$ *is the denominator of* τ^**, and primes denote derivatives with respect to* γ*.*

Under the simplifying assumption that welfare weights are approximately stable in γ *(*$(\frac{\Sigma}{N})' \approx \bar{g}' \approx 0$*), the sign of* $\partial\tau^*/\partial\gamma$ *is determined by the sign of* E'*: if the efficiency cost shrinks as* γ *rises (*$E' > 0$*), then* $\partial\tau^*/\partial\gamma > 0$*.*

The derivative of τ^* with respect to transferability has three terms. The first captures how the efficiency cost E responds to γ. The other two capture changes in the net social value of the benefit, $(\frac{\Sigma}{N})'$, and the taxpayer welfare weight $\bar{g}'$. I assume both are small, a simplifying assumption that welfare weights are stable as γ varies. Under it, the sign of $\partial\tau^*/\partial\gamma$ reduces to the sign of E'.

Now to see what determines the sign of E' and thus the sign of $\partial\tau^*/\partial\gamma$, recall that $E = \epsilon_m(\alpha_m - \beta_m) + \epsilon_f(\alpha_f - \beta_f)$. The elasticity ϵ_i depends on the density of households at two margins:

$$\epsilon_i \;=\; -\frac{(1-\tau)\, z_i}{F_i}\Big[\; \underbrace{q_i(T_i)\,[1 - Q_j(T_j)]}_{\text{FF margin: FF}\to\text{one-formal}} \;+\; \underbrace{q_i(T_i^{\text{low}})\, Q_j(T_j^{\text{low}})}_{\text{II margin: one-formal}\to\text{II}} \;\Big]. \tag{29}$$

The first term captures couples at the upper threshold T_i where the second spouse considers exiting formality while the partner remains formal. The second term captures couples at the lower threshold T_i^{low} where the sole provider considers exiting while the partner is already informal. As γ rises, the two thresholds move in opposite directions: T_i rises by b and T_i^{low} falls by b, and therefore the threshold gap $T_i - T_i^{\text{low}} = 2\gamma b$ widens. This is the same mechanism described in the introduction: higher γ makes one-formal arrangements more appealing from both extremes. So there are two channels at play here:

- The provider-role channel (II margin): higher γ raises the exit cost to $(1+\gamma)b$, making the sole provider *less* willing to leave one-formal. That is $E' < 0$
- The second-spouse channel (FF margin): higher γ shrinks the exit cost from $(1-\gamma)b$, making the second spouse *more* willing to leave FF. That is, $E' > 0$.

The same two forces also appear in β_i, the coverage-loss ratio. The weight in β_i decomposes as $(1 - \gamma + 2\gamma\mu_i) = (1-\mu_i)(1-\gamma) + \mu_i(1+\gamma)$: a weighted average of the FF-margin coverage loss $(1-\gamma)$ and the II-margin coverage loss $(1+\gamma)$. As γ rises, the FF-margin loss shrinks (pushing β_i down, raising $|E|$) while the II-margin loss grows (pushing β_i up, shrinking $|E|$). The net direct effect on the weight is $-1 + 2\mu_i$, which flips sign at $\mu_i = 1/2$. Because $E = \epsilon_i(\alpha_i - \beta_i)$ combines the elasticity (how many people move) with the net fiscal impact per mover $(\alpha_i - \beta_i)$, and μ_i governs how both respond to γ — dampening the elasticity and raising β_i when $\mu_i > 1/2$, amplifying the elasticity and lowering β_i when $\mu_i < 1/2$ — the two effects reinforce each other, making $\mu_i = 1/2$ a sharp threshold.

Proposition 3 (Two regimes)**.** *The sign of* E' *depends on the relative mass at the FF and II margins, summarized by* μ_i*, the share of spouse* i*'s marginal exiters at the II margin:*

$$\mu_i \;=\; \frac{q_i(T_i^{low})\, Q_j(T_j^{low})}{q_i(T_i)\,[1 - Q_j(T_j)] \;+\; q_i(T_i^{low})\, Q_j(T_j^{low})}.$$

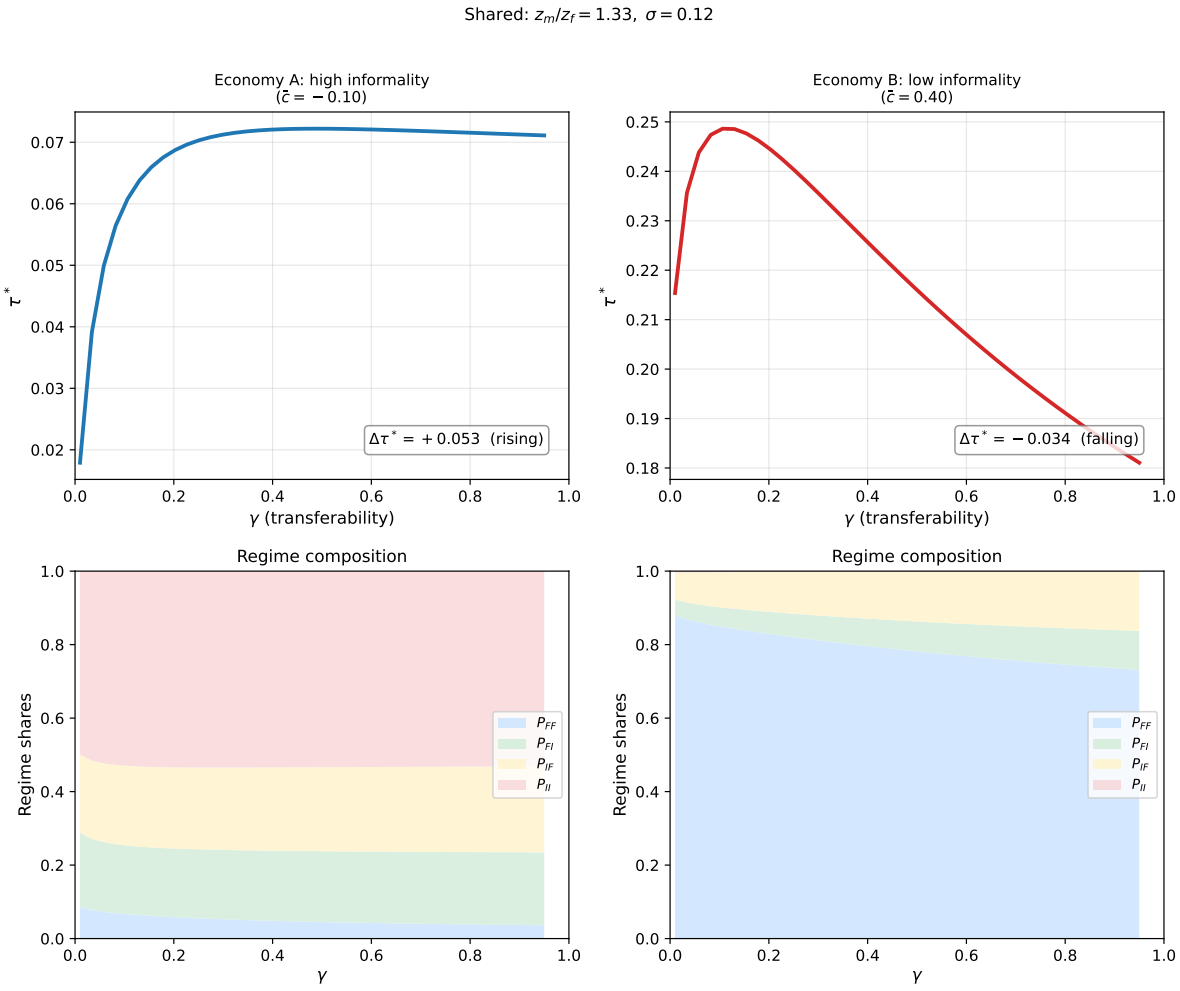
When $\mu_i > 1/2$ *(II margin dominant), the provider-role channel dominates:* $E' > 0$ *and* $\partial\tau^*/\partial\gamma > 0$*. When* $\mu_i < 1/2$ *(FF margin dominant), the second-spouse channel dominates:* $E' < 0$ *and* $\partial\tau^*/\partial\gamma < 0$*.*

In economies with substantial one-formal households, i.e., middle-income economies with non-trivial informality, the provider-role channel is expected to dominate. Raising γ makes the sole provider less elastic faster than it makes the second spouse more elastic, the efficiency cost shrinks, and the planner's best response is a higher tax. In fully-formal economies, i.e., developed economies where the one-formal mass is thin, the second-spouse channel is expected to dominate. The efficiency cost grows with γ, and the planner's best response is a lower tax.

Numerical illustration. Figure 2 illustrates the two regimes using a toy simulation of the model. Both economies share the same primitives: a wage ratio $z_m/z_f = 1.33$ ($z_m = 1.2$, $z_f = 0.9$), no female subsidy ($s = 0$), and stigma costs drawn from a logistic distribution with scale $\sigma = 0.12$. The welfare weights are $g^{FF} = 0.3$, $g^{FI} = g^{IF} = 1.3$, $g^{II} = 2.0$, so the planner values informal households more than formal ones. The only parameter that differs is the center of the stigma distribution $\bar{c}$, which governs the baseline level of informality: Economy A has $\bar{c} = -0.10$ (low stigma costs, high informality) and Economy B has $\bar{c} = 0.40$ (high stigma costs, low informality).

In Economy A, roughly half of all couples are in regime II, $\mu_i > 1/2$, and τ^* rises by 5.3 percentage points as γ goes from 0 to 1 — the provider-role channel dominates. In Economy B, over 80% of couples are in FF, $\mu_i \approx 0$, and τ^* falls by 3.4 percentage points — the second-spouse channel dominates. The bottom panels show the regime composition: Economy A has substantial mass in all four regimes, while Economy B is overwhelmingly FF. The regime switch is driven entirely by the labour-market structure ($\bar{c}$), not by preferences or wages.

Figure 2: Optimal $\tau^*(\gamma)$ and regime composition for two economies



The intuition is straightforward. In the high-informality economy, raising γ pulls both-informal couples into one-formal arrangements, because the provider role becomes valuable enough to justify the tax cost. The sole provider becomes less elastic, the efficiency cost of taxation shrinks, and the planner can set a higher τ. In the low-informality economy, raising γ pushes the second spouse in both-formal couples into informality, because there is less at stake from exiting when the partner already covers $(1+\gamma)b$. The second spouse becomes more elastic, the efficiency cost of taxation grows, and the planner's best response is a lower τ.

3.3. Optimal γ

Raising γ strengthens the provider role described in the introduction: the coverage the provider brings rises from b to $(1+\gamma)b$, while the second spouse's contribution falls from b to $(1-\gamma)b$. This pushes households toward one-formal arrangements from both extremes (FF $\to$ FI/IF and II $\to$ FI/IF), and these two margins have opposite effects on aggregate informality: the both-formal margin raises it, the both-informal margin lowers it. The net effect depends on which margin has more mass at the prevailing

policy.

The first-order condition for γ equates the coverage gain for one-formal households against the change in b for all beneficiaries:

$$\underbrace{b\big(P_{\text{FI}}\, g^{\text{FI}} + P_{\text{IF}}\, g^{\text{IF}}\big)}_{\text{coverage gain}} \;+\; \underbrace{\frac{db}{d\gamma}\left[2P_{\text{FF}}\, g^{\text{FF}} + (1+\gamma)(P_{\text{FI}}\, g^{\text{FI}} + P_{\text{IF}}\, g^{\text{IF}})\right]}_{\text{effect from change in } b} \;=\; 0. \tag{30}$$

The first term is the coverage gain: raising γ by $d\gamma$ gives each one-formal household an additional $b\, d\gamma$ in coverage, weighted by welfare weights g^{FI} and g^{IF}.

The second term captures the effect of the change in b for all covered households. Because $b = \tau Z/N$ is endogenous, raising γ changes the per-capita benefit. This change is weighted by $\bar{N} = N + \Sigma$, the welfare-weighted recipient mass. At the optimum, these two effects exactly offset.

The change in the per-capita benefit b can be decomposed as:

$$\frac{db}{d\gamma} \;\approx\; \frac{b}{N}\Big[-(P_{IF} + P_{FI}) \;-\; \tfrac{1}{\gamma}\left(\eta_m(\alpha_m - \beta_m) + \eta_f(\alpha_f - \beta_f)\right)\Big], \tag{31}$$

where the mechanical term comes from $b = \tau Z/N$ via $\partial N/\partial\gamma|_{P \text{ fixed}} = P_{FI} + P_{IF}$ (holding probabilities fixed). The corrected β_i from (24) (with μ_i in place of F_j) applies throughout.

The first term is the mechanical dilution of b: increasing γ expands the number of beneficiaries (N rises by $P_{\text{IF}} + P_{\text{FI}}$) without raising additional revenue, reducing b. The second term is the behavioral response: higher γ reshuffles households across formality regimes through the two margins described above. When the FF margin dominates ($\eta_i > 0$), more second spouses exit formality than first spouses enter from II, eroding the tax base and further reducing b. When the II margin dominates ($\eta_i < 0$), more households shift from II to one-formal than from FF to one-formal, expanding the tax base and partially offsetting the mechanical dilution.

The sign of η_i determines how b responds to higher γ, and therefore the nature of the optimal policy:

Case 1: $\eta_i > 0$ (both-formal margin dominates). The behavioral response adds to the mechanical dilution: strategic informality erodes the tax base, making b fall further. The resulting decline in b may offset the coverage gain at an interior point, so $\gamma^* \in (0, 1)$ may exist. The closed-form formula in Appendix B characterizes when this interior optimum obtains.

Case 2: $\eta_i < 0$ (both-informal margin dominates). The behavioral response works against the mechanical dilution: new formalizers expand the tax base, so b falls less than it would mechanically, or may even rise. The behavioral response moves workers toward formality rather than away from it. The first-order condition is strictly positive for every $\gamma \geq 0$ because the coverage gain is never offset by the behavioral channel – the behavioral channel now reinforces it. The welfare function is monotonically increasing on $[0, 1]$, placing the optimum at the upper corner at $\gamma = 1$.

Note that the empirical strategy in Section 4 identifies ϵ_i and ϕ_i (the response to the tax rate and the cross-spousal feedback) under a fixed $\gamma = 1$, but does not directly estimate η_i, since there is no observed variation in transferability. The sign of η_i is therefore not pinned down by the data. However, the presence of strategic substitution ($\phi < 0$) is consistent with the both-formal margin being active, suggesting that $\eta_i > 0$ is the empirically plausible case in this setting.

3.4. Elasticity estimation

The key extensive-margin elasticities with respect to the net-of-tax rate are defined as:

$$\epsilon_m = \frac{dF_m}{d(1-\tau)}\frac{1-\tau}{F_m} \quad \text{and} \quad \epsilon_f = \frac{dF_f}{d(1-\tau)}\frac{1-\tau}{F_f}, \tag{32}$$

where F_m and F_f are the marginal informality probabilities of (13). This reflects the fact that the planner chooses a single payroll tax rate, τ, for all workers. Throughout this subsection, ϵ_i refers to the elasticity of marginal informality with respect to the net-of-tax rate, treating the diagonal channel as negligible (as discussed in Section 3).

In the joint household model, the marginal informality probability of spouse i ($F_i = P_{iF} + P_{iI}$) depends on the net-of-tax rate $1-\tau$ and indirectly on the partner's marginal informality F_j through joint decision-making. By the chain rule, the total derivative of F_i with respect to $1-\tau$ decomposes as

$$\frac{dF_i}{d(1-\tau)} = \frac{\partial F_i}{\partial(1-\tau)} + \frac{\partial F_i}{\partial F_j}\frac{dF_j}{d(1-\tau)}. \tag{33}$$

The first term represents the direct effect—the sensitivity of i's informality to changes in $1-\tau$ holding j's response constant—while the second term captures the indirect (cross-spousal) effect arising from the fact that a change in $1-\tau$ also alters F_j, which in turn influences F_i through the joint decision rule. I define the cross-spousal elasticity as

$$\phi_i = \frac{\partial F_i}{\partial F_j}\cdot\frac{F_j}{F_i}, \tag{34}$$

so that the overall elasticity for i can be expressed as

$$\epsilon_i = \frac{dF_i}{d(1-\tau)}\cdot\frac{1-\tau}{F_i} = \epsilon_i^{\text{direct}} + \phi_i\,\epsilon_j, \tag{35}$$

where $\epsilon_j = \frac{dF_j}{d(1-\tau)}\cdot\frac{1-\tau}{F_j}$ is the spouse elasticity defined with respect to $1-\tau$.

So far, the presence of the subsidy for women has been ignored. This subsidy is exogenously given and not a choice variable. As a result, the effective payroll tax rate for women is:

$$\tau_f = (1-s)\tau \tag{36}$$

and the corresponding effective net-of-tax rate for women is $1-\tau_f = 1-(1-s)\tau$. Given this, to capture the true behavioral response of women, it is more appropriate to define the elasticity with respect to the effective net-of-tax rate, since women respond to their actual effective tax burden in the economy:

$$\tilde{\epsilon}_f = \frac{dF_f}{d\left[1-(1-s)\tau\right]}\cdot\frac{1-(1-s)\tau}{F_f}. \tag{37}$$

One can rewrite the effective-tax elasticity as:[8]

[8]Denote $u = 1-\tau$ and $v = 1-(1-s)\tau$. Then $v = s + (1-s)u$. Differentiating v with respect to u gives $dv/du = 1-s$. Applying the chain rule, $dF_f/dv = dF_f/du \cdot du/dv = \frac{1}{1-s}dF_f/du$. Thus, the effective elasticity is

$$\tilde{\epsilon}_f = \frac{1-(1-s)\tau}{(1-s)(1-\tau)}\,\epsilon_f \tag{38}$$

Thus, to convert the elasticity defined with respect to the nominal net-of-tax rate (ϵ_f) into the one defined with respect to the effective net-of-tax rate ($\tilde{\epsilon}_f$), one multiplies by the factor $\frac{1-(1-s)\tau}{(1-s)(1-\tau)}$. Equivalently, one can express $\epsilon_f = \frac{(1-s)(1-\tau)}{1-(1-s)\tau}\,\tilde{\epsilon}_f$, which implies that $\epsilon_f < \tilde{\epsilon}_f$ whenever $s > 0$.

This distinction is important because ϵ_f—the elasticity with respect to the common statutory net-of-tax rate—is the relevant object that enters the optimal payroll tax formula. Even though women may exhibit a large behavioral response to their effective tax rate ($\tilde{\epsilon}_f$), from the planner's perspective, the efficiency cost of taxing women is captured by ϵ_f, which is smaller due to the subsidy. Intuitively, the subsidy protects women from the full statutory distortion, lowering the effective burden they experience relative to the statutory tax rate, and thus reducing their contribution to the marginal efficiency cost of raising payroll taxes.

Coming back to the decomposition of elasticities, one can write the elasticities for men and women as:

$$\epsilon_m = \epsilon_m^{\text{direct}} + \phi_m\,\epsilon_f, \qquad \epsilon_f = \epsilon_f^{\text{direct}} + \phi_f\,\epsilon_m \tag{39}$$

where $\epsilon_m^{\text{direct}}$ and $\epsilon_f^{\text{direct}}$ denote the direct elasticities of male and female informality with respect to the net-of-tax rate, and ϕ_m and ϕ_f capture cross-spousal effects (i.e., how changes in one spouse's formal status affect the other's incentives). Specifically, the female direct elasticity is related to the effective elasticity by:

$$\epsilon_f^{\text{direct}} = \frac{(1-s)(1-\tau)}{1-(1-s)\tau}\,\tilde{\epsilon}_f^{\text{direct}}. \tag{40}$$

In this setting, the female subsidy s reduces the effective tax rate for women to $(1-s)\tau$, directly affecting their informality response. The direct behavioral elasticity is captured by $\tilde{\epsilon}_f$, while the relevant elasticity for the optimal tax formula is $\epsilon_f^{\text{direct}}$, scaled down by the factor 1-s to reflect the reduced statutory burden. For men, the relevant elasticity is ϵ_m, which remains unaffected by s unless spillover effects across spouses exist. Any change in male informality—captured by ϕ_m—thus reflects the cross-spousal transmission of the subsidy's effect.

This decomposition allows one to separately identify the direct responsiveness of each spouse to changes in $1-\tau$ and the indirect effects transmitted through household dynamics, even though the planner's policy instrument remains the common payroll tax rate τ, with s treated as exogenous. Because payroll benefits are partially transferable within the household, the formality decision of one spouse affects the marginal value of formality for the other. When a spouse becomes formal and brings benefit coverage into the household, the incentive for the other spouse to seek formal employment is reduced. Thus, the cross-spousal elasticities ϕ_m and ϕ_f capture this substitutability in formality decisions driven by the transferability of benefits.

$\tilde{\epsilon}_f = \frac{dF_f}{dv}\cdot\frac{v}{F_f} = \frac{1}{1-s}\frac{dF_f}{du}\cdot\frac{v}{F_f}$. On the other hand, the elasticity with respect to the nominal net-of-tax rate is $\epsilon_f = \frac{dF_f}{du}\cdot\frac{u}{F_f}$. Therefore, the relation between the two elasticities is $\tilde{\epsilon}_f = \frac{v}{(1-s)u}\epsilon_f$.

Two remarks on the structure of this system. First, the decomposition is simultaneous: ϵ_m depends on ϵ_f and vice versa. Solving the system yields the reduced-form elasticities

$$\epsilon_m = \frac{\epsilon_m^{\text{direct}} + \phi_m \, \epsilon_f^{\text{direct}}}{1 - \phi_m \phi_f}, \qquad \epsilon_f = \frac{\epsilon_f^{\text{direct}} + \phi_f \, \epsilon_m^{\text{direct}}}{1 - \phi_m \phi_f},$$

which requires $\phi_m \phi_f \neq 1$. Since the cross-spousal effects reflect strategic substitution ($\phi_m, \phi_f < 0$ under the free-riding channel), the product $\phi_m \phi_f > 0$, and the condition $\phi_m \phi_f < 1$ ensures that the feedback loop between spouses is dampened rather than explosive.

Second, the partial derivative $\partial F_i / \partial F_j$ is well-defined in this model because the subsidy s provides an instrument that shifts F_f without directly affecting men's thresholds. Along the variation induced by s, the cross-spousal elasticity ϕ_m can be identified as the ratio of the male to female response. This is the object recovered in the empirical section below.

The BTM-based estimation in Section 4 identifies the effective-rate elasticity $\tilde{\epsilon}_f$ and the cross-spousal feedback ϕ_m. Converting these to the formula objects requires two steps: (i) rescaling $\tilde{\epsilon}_f \to \epsilon_f^{\text{direct}}$ via the subsidy factor in (38), then (ii) solving the linear system above for ϵ_m and ϵ_f, the total elasticities that enter the optimal-tax formula (27).

4. Empirics

This section exploits the introduction of a subsidy to formal employment in Chile to estimate the informality elasticity for women (ϵ_f) and the cross-elasticity between spouses (ϕ_m). The empirical evidence leads with the administrative panel, which delivers the main results—the provider-role and free-riding margins identified within households—and then draws on the CASEN household survey for the two things the administrative data cannot supply: a multi-year pre-period to test parallel trends, and a direct measure of informality.

4.1. The Subsidy: Bono al Trabajo de la Mujer (BTM)

The BTM program was launched in Chile in 2012 following the enactment of Law No. 20.595, as a policy response to the persistently low rates of formal employment among women. It provides wage subsidies to both employees and employers conditional on the formal employment of women. As such, the program functions as a conditional cash transfer, with eligibility contingent upon maintaining formal employment. Formal employment is defined as having a signed contract with an employer registered in the Chilean Internal Revenue Service and making the required social security contributions.

Eligibility for BTM is determined through a continuous targeting score based on information from the Social Protection Form (Ficha de Protección Social, FPS) and household income data provided by SENCE. At launch in 2012, eligible women were those with a score of 98 points or lower, corresponding to the 30 percent most socioeconomically vulnerable women within the eligible age group. In January 2014, coverage was expanded to the 35 percent most vulnerable (score cutoff of 104), and in January 2015 to the 40 percent most vulnerable (score cutoff of 113).

The BTM benefits both the female worker and her employer. To qualify, the worker must be between

25 and 60 years of age, formally employed, and belong to the 30, 35, or 40% most vulnerable socioeconomic strata (depending on the year), as determined by FPS scores and household income per capita. The amount of the subsidy depends on the worker's gross earnings, providing a monthly benefit of up to 20% of the wage for the worker and up to 10% for the employer. Specifically:

- Phase-in (CLP 0 to 170,000): The worker receives 20% of her gross income; the employer receives 10% of the worker's gross income.
- Constant benefit (CLP 170,000 to 210,000): The worker receives a fixed transfer of CLP 34,000; the employer receives CLP 17,000.
- Phase-out (CLP 210,000 to 380,000): The worker receives CLP 34,000 minus 20% of the difference between her gross income and CLP 210,000; the employer receives CLP 17,000 minus 20% of that difference.

The maximum benefit is CLP 34,000 for the worker and CLP 17,000 for the employer, which accounts for up to 20% of the wage for the worker and up to 10% for the employer. See Figures 3 and 4.

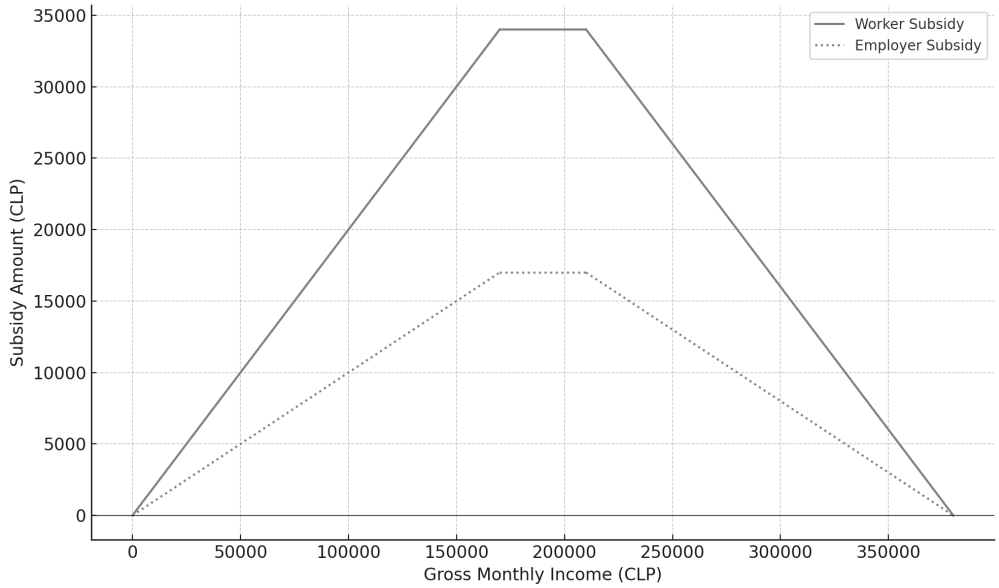


Figure 3: Monthly BTM Subsidy Amount by Gross Monthly Income

4.2. Primary Design: Baseline-Frozen Eligibility-DiD on Administrative Data

My primary evidence comes from the linked administrative panel underlying the BTM targeting system: the Ficha de Protección Social (FPS, 2010–2015), extended through the Registro Social de Hogares (RSH, 2016–2022), comprising 97.5 million individual-year records. Formal employment is observed directly from social-security contribution records, and spouses are linked within the family unit—identified by the household–*puntaje* pair, since sub-families within a single dwelling carry distinct targeting scores. After restricting to individuals aged 25–60 observed at baseline, the estimation sample contains roughly

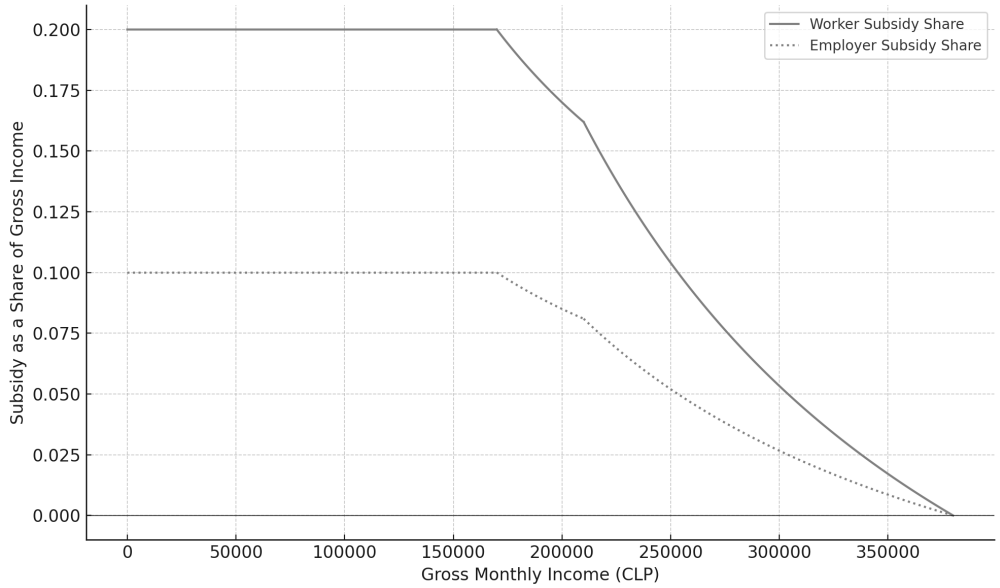


Figure 4: BTM Subsidy as a Share of Gross Monthly Income

30 million person-years for women and 30 million for men. The headline estimates use the 2010–2015 FPS window; the 2016–2022 RSH extension is reserved for robustness because of the FPS-to-RSH measurement break.

Two features make this design cleaner than the survey-based ITT below. First, the administrative panel follows the same individuals over time, so eligibility and spousal formality can be *frozen at their pre-policy* (2010–2011) values rather than measured contemporaneously—contemporaneous spousal status is itself an outcome of the policy. Second, the panel is large enough to identify the household-composition interactions that are the structural objects of the model. I define $\text{Elig}_i = 1$ for women meeting baseline eligibility (female, age 25–60, baseline *puntaje* $\leq$ 11,734), $\text{SpInf}_i = 1$ for women whose husband had no formal contract at baseline, and $\text{Post}_t = \mathbf{1}[t \geq 2012]$, since BTM payments began in the fourth quarter of 2012.

Block 1: Average effect on eligible women. The pooled eligibility-DiD,

$$Y_{it} = \alpha_i + \lambda_t + \beta\,(\text{Elig}_i \times \text{Post}_t) + \varepsilon_{it}, \tag{41}$$

estimated on all women aged 25–60 observed at baseline, yields $\beta = +1.62$ percentage points (s.e. 0.03; $N = 18.6$M). The event study is flat before 2012—the 2010 pre-coefficient is -0.14pp—and jumps sharply at the statute year, reaching $+2.25$pp by 2015. I interpret β as the effect of bottom-40 eligibility.

Block 2: Provider-role differential. The model's sharp prediction is not about the level but about *which* households respond. I estimate the triple-difference

$$Y_{it} = \alpha_i + \lambda_t + \tau\,(\text{Elig}_i \times \text{Post}_t) + \delta\,(\text{Elig}_i \times \text{Post}_t \times \text{SpInf}_i) + \theta\,(\text{SpInf}_i \times \text{Post}_t) + \varepsilon_{it} \tag{42}$$

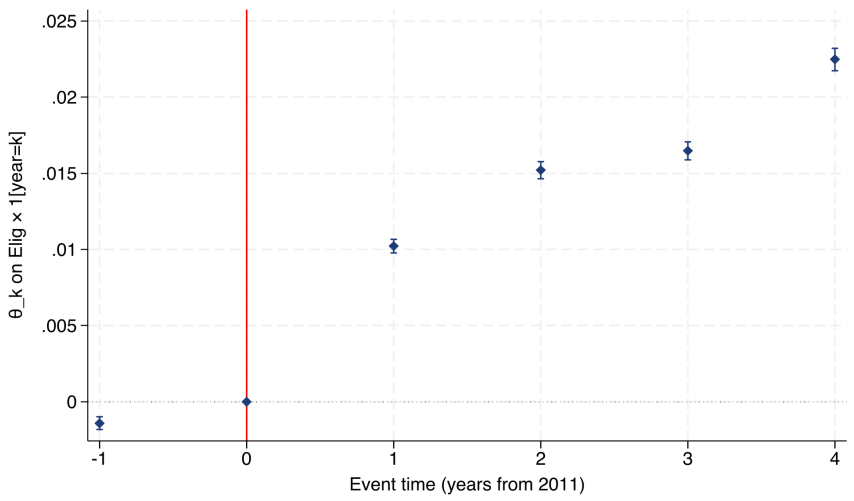


Figure 5: Block 1: eligibility-DiD event study, women's formal employment. Flat before 2012, jumping at the statute year.

on partnered women with observable baseline husband formality. Here τ is the effect on eligible women married to a *formal* husband at baseline, and δ—the provider-role differential—is the additional response of eligible women married to an *informal* husband. I estimate $\tau = +1.48$pp (s.e. 0.05) and $\delta = +0.34$pp (s.e. 0.08), so eligible women with informal husbands formalize by $\tau + \delta = +1.81$pp. The pre-policy differential is essentially zero, and δ_k grows monotonically from +0.26pp in 2012 to +0.54pp by 2015. A positive δ is exactly the provider-role prediction: when the household lacks formal coverage through the husband, the eligible wife is more likely to become the formal "coverage provider." Importantly, any policy enacted alongside the BTM differences out of δ as long as it affects both husband-formality groups equally.

Block 3: Husband free-riding. The complementary prediction concerns husbands. I estimate the reduced form

$$H_{jt} = \alpha_j + \lambda_t + \rho\,(\text{WifeElig}_j \times \text{Post}_t) + \varepsilon_{jt}, \tag{43}$$

where WifeElig_j is the husband's exposure through his wife's baseline eligibility, splitting the sample by the husband's *own* baseline formality. Among husbands who held a formal job at baseline, wife eligibility reduces formal employment by $\rho_F = -2.14$pp (s.e. 0.06), comparing 2012–2015 with 2010–2011—husbands withdraw from formal work once coverage is secured through the wife. Because the formal-husband series already moves before 2012, I net out a placebo estimated on *unpartnered* vulnerable men, for whom the spillover channel cannot operate. The correction uses the 2015 event-study coefficients, both relative to 2011: formal husbands fall by 1.86pp (Figure 7) and the unpartnered men by 1.44pp. Relative to 2011, the placebo-corrected free-rider effect is therefore $-1.86 - (-1.44) \approx -0.42$pp: most of the raw decline is secular, but the residual is in the predicted direction and of the same order of magnitude as the provider-role differential δ. This correction depends on the reference year, because the two groups move in opposite directions before the policy. The 2010 coefficient is +0.94pp for formal

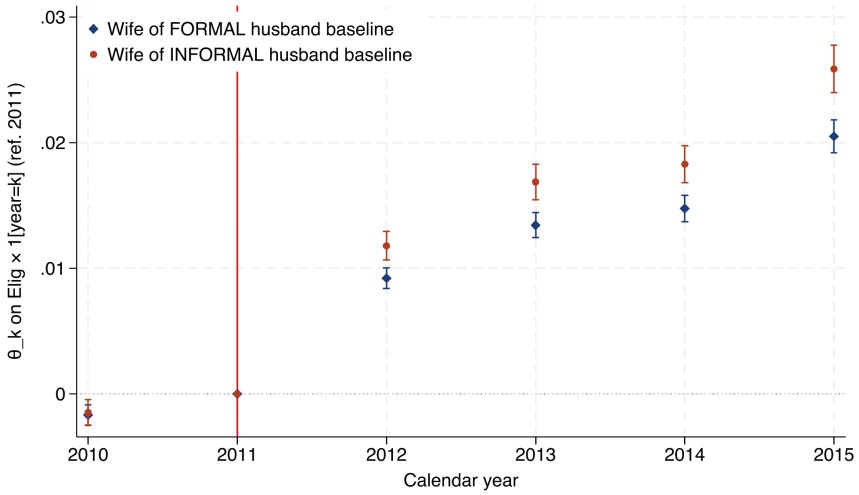


Figure 6: Block 2: eligible women's formal employment, split by husband's baseline formality. The informal-husband series (provider-role margin) runs above the formal-husband series; the gap δ_k widens to +0.54pp by 2015.

husbands and −1.21pp for the unpartnered men, a gap of 2.15pp (s.e. 0.11) that is larger than any post-2012 gap (0.24 to 0.42pp). Measured against 2010 instead, the corrected effect is about −2.6pp, and the placebo explains little of the raw decline. With only two pre-policy years, the data cannot choose between the two baselines; under both, the free-rider effect has the predicted sign.

Reading. Together, Blocks 2 and 3 identify the model's two key behavioral margins—the provider-role differential $\delta > 0$ and the free-rider response $\rho_F < 0$—off baseline household composition, without relying on the pooled level effect. This is the core empirical contribution: direct, within-household evidence of specialization into a single formal "coverage provider" when social-insurance benefits are transferable across spouses. The survey evidence below validates this design where the administrative panel is weakest and confirms its findings under a direct measure of informality.

4.3. Survey Evidence: Pre-Trends and Direct Informality

The administrative design has two blind spots that the CASEN household survey fills, and I use the survey for these two validation tasks only. First, because BTM was enacted in 2012 and the administrative panel begins in 2010, the within-person estimates rest on a single clean pre-treatment year; CASEN, fielded biennially since 2006, supplies the multi-year pre-period needed to test parallel trends. Second, the administrative records observe only formal-sector registration, so informality appears there only as withdrawal from the formal force; CASEN measures informality directly, through pension affiliation and written contracts. The survey cannot, however, re-identify the structural margins of Blocks 2 and 3: as a repeated cross-section it cannot freeze spousal formality at a baseline year, so a split on contemporaneous husband status would be endogenous to the policy.

The survey design compares households in the bottom four income deciles (treated, broadly BTM-

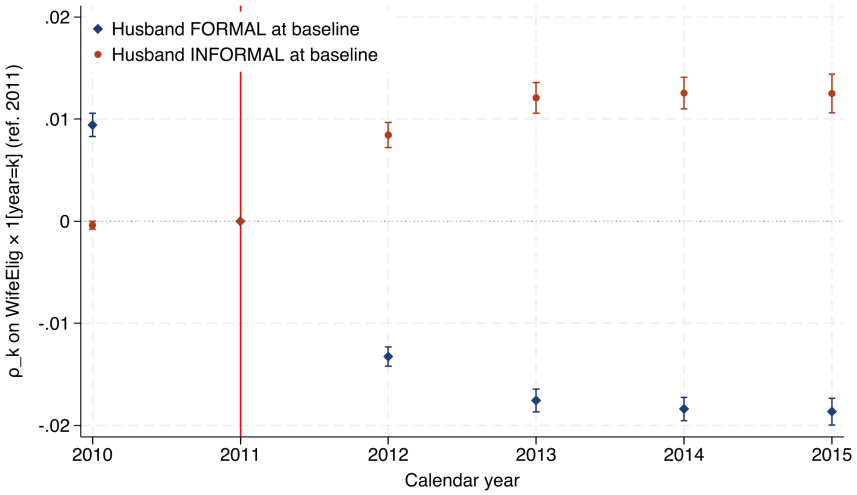


Figure 7: Block 3: husbands' formal employment, split by own baseline formality. Formal-at-baseline husbands of eligible wives drop sharply after 2012 (free-riding); already-informal husbands track the secular trend.

eligible) to those in deciles five through seven—an economically similar but ineligible control group—in an event study referenced to 2011, with standard errors clustered by region.

Pre-trends. Figure 8 plots the event-study coefficients on formal employment for both genders. The pre-treatment coefficients (2006 and 2009) are statistically indistinguishable from zero for women and men alike, while the post-2011 coefficients are large and precisely estimated. This is exactly the parallel-trends evidence the single administrative pre-period cannot provide, and it supports the eligibility-DiD logic underlying the administrative estimates. The post-period coefficients are essentially unchanged by adding controls for age, education, and comuna fixed effects, so they are not a composition artifact.

The post-2011 coefficients are −5 to −7pp for women and −7 to −10pp for men. For men this is the same free-riding margin identified in the administrative panel—direct, survey-based confirmation that husbands shift toward informality. For women, the aggregate survey coefficient is *not* directly comparable to the within-person administrative estimate: a repeated cross-section captures between-group composition changes as well as individual transitions, so it should be read as evidence on the design's validity and on the men's margin, not as a second estimate of the female level effect.[9] Descriptively, the female decline concentrates entirely in households whose husband is *currently* informal (Figure 9), consistent with the joint-informality margin the model predicts—though contemporaneous husband status is itself endogenous, so this is a descriptive partition rather than a heterogeneity estimate.

Direct informality. Because CASEN observes informality directly, I can confirm the findings are not an artifact of the formal-registration-only administrative measure. I report two definitions: a pension-

[9]The administrative Block 1 estimate is a within-person change for the same individuals over time; the CASEN coefficient is a change in the treated-minus-control gap, which also moves with the changing composition of who appears in the low-decile cell as the policy rolls out. The two answer different questions and need not share a sign.

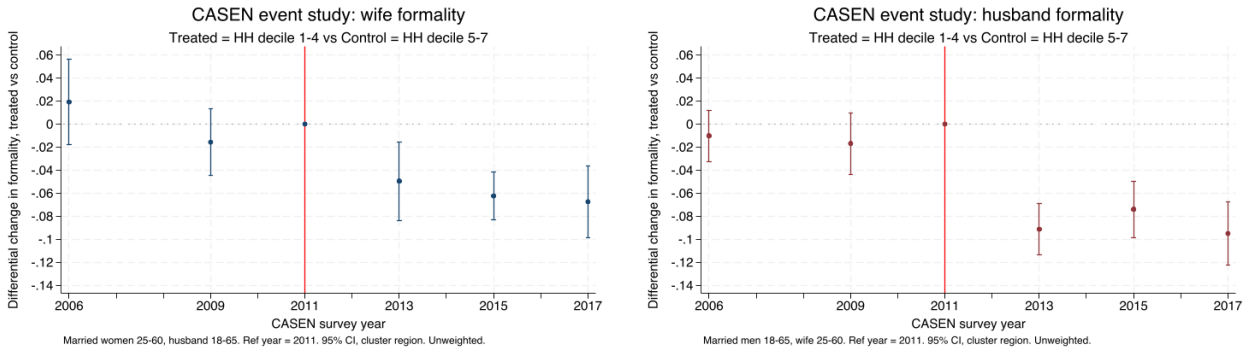


Figure 8: CASEN event study, formal employment, deciles 1–4 (treated) vs. 5–7 (control), referenced to 2011. Left: women. Right: men. Pre-2011 coefficients are insignificant for both genders; post-2011 coefficients are −5 to −10pp.

based one (informal = not affiliated to or not contributing to a pension system), whose question wording is stable across 2006–2017, and a written-contract one. Figure 10 shows that the post-policy rise in informality survives in 2015—the year least affected by a 2011 change in the contract question—under both definitions and for both genders, on the order of 7pp. The contract series is noisier in the pre-period because of that wording break, so I treat the pension-based definition as the cleaner headline; the point is that the direction and rough magnitude do not depend on the measure.

Scope. CASEN thus does two jobs and only two: it supplies the pre-period that defends parallel trends, and it confirms the men's free-riding margin under a direct measure of informality. It cannot identify the provider-role differential δ or the female level effect, which remain administrative-only.

4.4. Converting Estimates to Elasticities

To connect the empirical results to the theoretical model, I convert the ITT estimates into behavioral elasticities of informality with respect to the net-of-tax rate. The coefficients β^f and β^m entering the formulas below are the women's and men's administrative eligibility-DiD estimates (Blocks 1 and 3).

Recall that in the theoretical framework, extensive-margin elasticities are composed of two components: a direct effect of the net-of-tax rate on the individual, and an indirect effect that operates through the spouse:

$$\epsilon_i = \epsilon_i^{\text{direct}} + \phi_i \epsilon_j \tag{44}$$

The empirical estimates capture total elasticities, incorporating both the direct and cross-spousal responses.

For women, the estimated elasticity corresponds to the response to their *effective* net-of-tax rate, which reflects the presence of a subsidy s. This effective elasticity is defined as:

$$\tilde{\epsilon}_f = \frac{dF_f}{d[1-(1-s)\tau]} \cdot \frac{1-(1-s)\tau}{F_f} \tag{45}$$

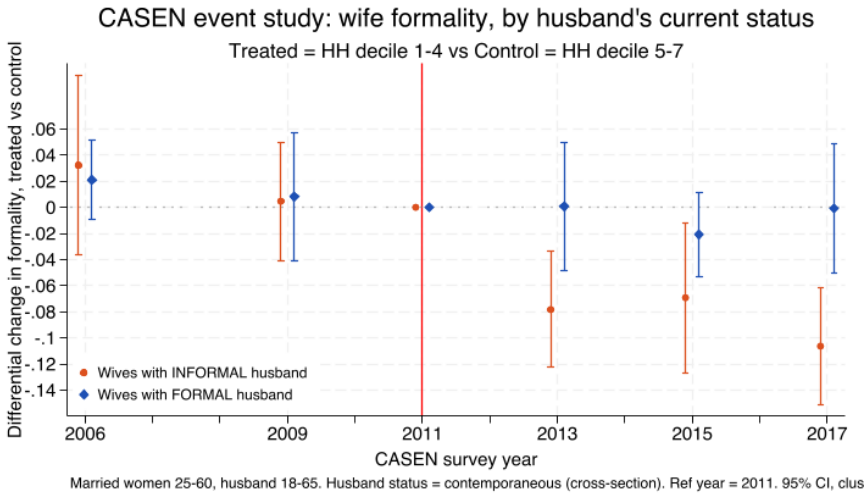


Figure 9: Women's CASEN event study split by husband's contemporaneous formality. The female decline concentrates in households with a currently-informal husband; wives of formal husbands show no post-2011 movement.

However, from the planner's perspective, what matters is the elasticity with respect to the *statutory* net-of-tax rate, since the optimal policy problem is defined over a single payroll tax rate τ. The scaling between effective and statutory elasticities derived in Section 3.3 applies to the *direct* component:

$$\epsilon_f^{\text{direct}} = \frac{(1-s)(1-\tau)}{1-(1-s)\tau} \cdot \tilde{\epsilon}_f^{\text{direct}} \tag{46}$$

This adjustment scales down the effective elasticity to reflect the fact that the planner observes only the statutory distortion, not the subsidized rate experienced by women. The total statutory elasticity is then $\epsilon_f = \epsilon_f^{\text{direct}} + \phi_f \, \epsilon_m$, where ϕ_f captures the feedback from male informality (see below for identification).

Empirically, the elasticity of female informality with respect to the effective net-of-tax rate is computed as:

$$\tilde{\epsilon}_f = \frac{\beta^f}{\Delta v} \cdot \frac{v}{\bar{F}_f} \tag{47}$$

where β^f is the ITT estimate for women, Δv is the change in the effective net-of-tax rate induced by eligibility, $v = 1-(1-s)\tau$, and $\bar{F}_f$ is the baseline informality rate for women.

For men, the statutory tax rate remains unchanged, so there is no direct effect from the BTM policy on male taxes. Instead, male informality responds only through the household-level spillover. Denoting the policy-specific male elasticity by ϵ_m^{BTM}, this implies:

$$\epsilon_m^{\text{BTM}} = \phi_m \cdot \epsilon_f \tag{48}$$

Note that this differs from the structural elasticity $\epsilon_m = \epsilon_m^{\text{direct}} + \phi_m \, \epsilon_f$ defined in Section 3.3, because here men's own tax rate does not change ($\epsilon_m^{\text{direct}}$ is not identified by this policy variation). The empirical elasticity estimated for men thus allows me to recover the *cross-spousal elasticity* ϕ_m, which is defined

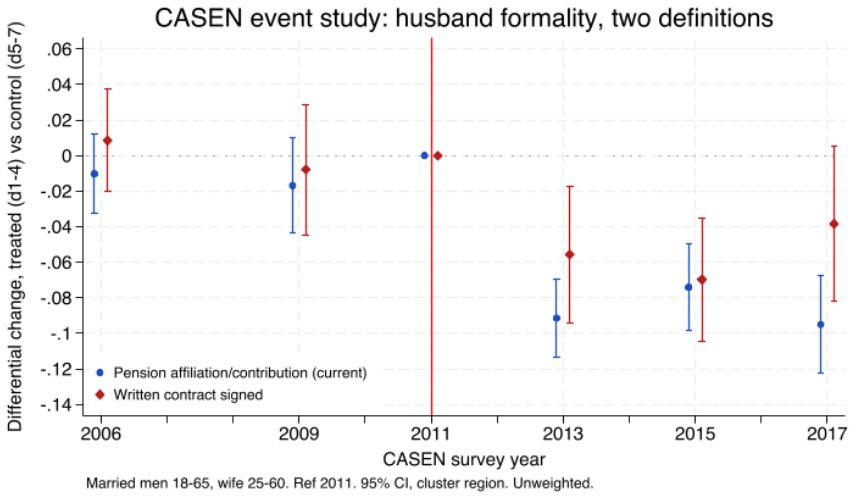


Figure 10: Men's CASEN event study under two informality definitions: pension-based (stable wording 2006–2017) and written-contract. The 2015 effect survives under both.

as:

$$\phi_m = \frac{\partial F_m}{\partial F_f} \cdot \frac{F_f}{F_m} \tag{49}$$

Finally, the cross-spousal elasticity is computed as:

$$\phi_m = \frac{\beta^m}{\beta f} \cdot \frac{\bar{F}_f}{\bar{F}_m} \tag{50}$$

where β^f is the ITT effect of eligibility on female informality, β^m is the ITT effect on male informality, and $\bar{F}_f, \bar{F}_m$ denote the baseline informality rates for women and men, respectively. This expression captures the degree to which changes in women's informality induced by the subsidy spill over to affect the informality decisions of their male partners.

Identification summary. The BTM variation identifies two objects: the total effective elasticity for women ($\tilde{\epsilon}_f$, from which $\tilde{\epsilon}_f^{\text{direct}}$ can be recovered) and the cross-spousal elasticity ϕ_m. Three structural parameters remain unidentified by this policy variation:

- $\epsilon_m^{\text{direct}}$: men's direct response to a change in the common payroll tax rate. Because men's statutory tax rate does not change under BTM, this parameter cannot be recovered. The baseline assumption used here is $\epsilon_m^{\text{direct}} = 0$, which implies $\epsilon_m = \phi_m \epsilon_f$. This may understate the structural male elasticity if men would also respond directly to a payroll tax change.

- ϕ_f: women's cross-response to changes in male informality. There is no exogenous variation in male formality independent of women's, so this parameter is not identified. Setting $\phi_f = 0$ (no feedback from men to women) yields $\epsilon_f = \epsilon_f^{\text{direct}}$, but this assumption rules out the symmetric channel from men to women.

- η_m, η_f: elasticities of informality with respect to transferability γ. Because γ does not vary in the Chilean setting, these are not identified by the BTM design. The optimal γ^* formula requires these objects, so calibrating γ^* requires additional assumptions or alternative sources of variation.

Appendix

A. Derivation of the first-order conditions

This appendix derives the first-order condition (26) from the Lagrangian. The same logic delivers the γ-FOC at the end.

Setup. The planner maximizes the Lagrangian

$$\mathcal{L}(\tau,\gamma) \;=\; \mathrm{SWF}(\tau,\gamma) \;+\; \lambda\big[\tau\, Z(\tau,\gamma,b) \;-\; b\, N(\tau,\gamma,b)\big], \tag{51}$$

where $b = b(\tau,\gamma)$ is implicitly determined by the balanced-budget constraint $\tau Z = bN$. Along the budget-constraint manifold, $\tau Z - bN \equiv 0$ as an identity in (τ,γ), so $d(\tau Z - bN)/d\tau = 0$ holds automatically. Hence

$$\frac{d\mathcal{L}}{d\tau} \;=\; \frac{d\,\mathrm{SWF}}{d\tau} \;+\; \lambda\cdot 0 \;=\; \frac{d\,\mathrm{SWF}}{d\tau}, \tag{52}$$

and the first-order condition reduces to $d\,\mathrm{SWF}/d\tau = 0$.

Differentiating the SWF. The social welfare function is

$$\mathrm{SWF}(\tau,\gamma) \;=\; \sum_{k\in\{FF,FI,IF,II\}} \iint_{\mathcal{R}_k} G\big(U^k(c_m,c_f)\big)\, q_m(c_m)\, q_f(c_f)\, dc_m\, dc_f.$$

Differentiating in τ produces two kinds of terms: interior changes because U^k depends on τ (directly, and through b), and boundary changes because the regions $\mathcal{R}_k$ move with τ. By the envelope condition, $U^k = U^{k'}$ at every regime boundary (that is how the boundaries are defined), so $G(U^k) = G(U^{k'})$ along each boundary and the boundary contributions cancel. Only the interior change matters:

$$\frac{d\,\mathrm{SWF}}{d\tau} \;=\; \sum_k \iint_{\mathcal{R}_k} G'(U^k)\left[\frac{\partial U^k}{\partial \tau} \;+\; \frac{\partial U^k}{\partial b}\cdot\frac{db}{d\tau}\right] f_m\, f_f\, dc_m\, dc_f. \tag{53}$$

Partial derivatives of regime utilities. Read off equations (5)–(8):

Regime k	$\partial U^k/\partial\tau$	$\partial U^k/\partial b$
FF	$-\,[z_m + (1-s)z_f]$	2
FI	$-\,z_m$	$1+\gamma$
IF	$-\,(1-s)z_f$	$1+\gamma$
II	0	0

Welfare weights and substitution. Define the regime welfare weights g^k by

$$\iint_{\mathcal{R}_k} G'(U^k)\, f_m f_f\, dc_m dc_f \;=\; P_k\, g^k\, \lambda,$$

i.e., g^k is the average marginal social value of consumption in regime k, normalised by the Lagrange multiplier λ. Substituting into (53) and grouping the $\partial U^k/\partial\tau$ and $\partial U^k/\partial b$ terms:

$$\begin{aligned}\frac{d\,\mathrm{SWF}}{d\tau} &= -\lambda\left[(z_m+(1-s)z_f)\,P_{FF}g^{FF}+z_m\,P_{FI}g^{FI}+(1-s)z_f\,P_{IF}g^{IF}\right]\\ &\quad+\lambda\,\frac{db}{d\tau}\left[2\,P_{FF}g^{FF}+(1+\gamma)\,P_{FI}g^{FI}+(1+\gamma)\,P_{IF}g^{IF}\right]\\ &= -\lambda\,Z\,\bar{g}\;+\;\lambda\,\frac{db}{d\tau}\,\bar{N},\end{aligned}$$

using the definitions

$$\bar{g}=\frac{(z_m+(1-s)z_f)P_{FF}g^{FF}+z_mP_{FI}g^{FI}+(1-s)z_fP_{IF}g^{IF}}{Z},\qquad \bar{N}=2P_{FF}g^{FF}+(1+\gamma)\,(P_{FI}g^{FI}+P_{IF}g^{IF}).$$

$\bar{g}$ is the income-weighted average welfare weight of formal workers; $\bar{N}$ is the welfare-weighted recipient mass. Setting $d\,\mathrm{SWF}/d\tau=0$ and dividing by λ delivers the τ-FOC (26):

$$-Z\,\bar{g}\;+\;\frac{db}{d\tau}\,\bar{N}\;=\;0.$$

The γ-FOC. The same logic applies to differentiation in γ. The partial derivatives are: $\partial U^{FF}/\partial\gamma=0$, $\partial U^{FI}/\partial\gamma=b$, $\partial U^{IF}/\partial\gamma=b$, $\partial U^{II}/\partial\gamma=0$, and the $\partial U^k/\partial b$ derivatives are the same as in the table above. Substituting and simplifying yields

$$b\,(P_{FI}\,g^{FI}+P_{IF}\,g^{IF})\;+\;\frac{db}{d\gamma}\,\bar{N}\;=\;0,$$

which is the γ-FOC stated in Section 3.2.

B. Closed-form expression for γ^*

Substituting the decomposition of $db/d\gamma$ from (31) into the first-order condition for γ and solving yields:

$$\gamma^*\;=\;\frac{\left(1+\dfrac{\Sigma}{N}\right)\left[\eta_m(\alpha_m-\beta_m)+\eta_f(\alpha_f-\beta_f)\right]}{\Omega-\left(1+\dfrac{\Sigma}{N}\right)(P_{\mathrm{FI}}+P_{\mathrm{IF}})},\tag{54}$$

where $\Omega\equiv P_{\mathrm{FI}}\,g^{\mathrm{FI}}+P_{\mathrm{IF}}\,g^{\mathrm{IF}}$ is the welfare-weighted mass of one-formal households (the direct beneficiaries of coverage), Σ/N is the per-recipient net social value of benefits defined in Section 3.1, and β_i uses the corrected μ_i from (24).

The numerator captures the behavioral cost of transferability, scaled by the social value of benefits. The denominator captures the net coverage gain: the direct welfare gain Ω minus the cost of mechanical dilution, also scaled by $1+\Sigma/N$. When $\eta_i>0$ (both-formal margin dominates), the numerator is positive and an interior $\gamma^*\in(0,1)$ may exist whenever the denominator is also positive. When $\eta_i<0$ (both-informal margin dominates), the numerator is negative, the numerator of (52) is negative, so the unconstrained formula suggests $\gamma^*<0$. But this reflects the fact that the behavioral response now

reduces rather than increases informality: $\partial W/\partial\gamma > 0$ for every $\gamma \geq 0$, because the coverage gain is reinforced rather than offset by the behavioral channel. The welfare function is therefore monotonically increasing on $[0, 1]$, and the constrained optimum is the upper corner $\gamma = 1$. The empirically plausible case for Chile is $\eta_i > 0$ (consistent with strategic substitution $\phi < 0$).

Note that the closed-form also exhibits a counterintuitive comparative static: a larger $|\eta_i|$ when $\eta_i > 0$ increases the numerator, pushing γ^* upward rather than downward. This arises because η_i enters both the behavioral cost (numerator) and the benefit dilution that determines how quickly b falls (implicitly, through $db/d\gamma$ in the denominator). The FOC presentation in the main text avoids this issue by keeping the two channels separate.

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